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2 January 1985
SUPERSEDING
MIL-F-80188A
2 May 1974

MILITARY SPECIFICATION

FURNACES, HEAT TREATING, ELECTRIC, METAL LINED BOX DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers natural atmosphere heat treating furnaces of the metal lined, electrically heated, horizontal box design. The furnaces are normally referred to as industrial ovens by manufacturers.

1.2 Classification.

1.2.1 Types and classes. The furnaces shall be of the following types and classes as specified (see 6.2.1).

Type I - Gravity Convection

Class A - 200 - 1000°F Operating Temperature Range
* Class B - 300 - 1350°F Operating Temperature Range

Type II - Forced Convection

Class A - 200 - 1000°F Operating Temperature Range
* Class B - 300 - 1350°F Operating Temperature Range

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3424

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* 1.2.2 Sizes. The size of the furnace chamber required shall be as specified (see 6.2.1), and should be selected from the sizes listed in table I. The dimensions specified for each size shall define the minimum inside dimensions of the heating chamber and if a furnace with a larger chamber is offered, the dimensions shall not exceed the dimensions specified by more than 25 percent. The door opening clearance shall be not less than the width and height dimension specified for the size furnace.

TABLE I. Furnace chamber dimensions (inches).

Size	Width	Length	Height
1	18	24	18
2	18	36	18
3	24	36	24
4	24	48	24
5	30	60	30
6	30	48	30
7	36	60	36
8	36	72	36
9	42	72	42

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-3296 - Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking) Packaging of.

MIL-H-6088 - Heat Treatment of Aluminum Alloys.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NEMA ICS 1 - Industrial Control and Systems.

ANSI/MC96.1 - Temperature Measurement Thermocouples.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.
ASTM-E-380 - Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

* 3.2 Design. The furnace shall be new and one of the manufacturer's current models. The furnace shall be of tne horizontal box design with a single opening in the end of the chamber for loading and unloading the work in a horizontal plane. The furnace chamber shall be a double wall metal construction with insulating material separating the inner and outer wall. Heating elements shall be separated from the work chamber by baffles, preventing transfer of heat to the work by radiation. Chamber design shall provide for the heated air to enter the work chamber through ducts or orifice perforated inner walls, and flow through the work area in a pattern that will obtain the temperature and uniformity specified herein. Each chamber shall be provided with fresh air intake and exhaust ports with means to manually adjust the port openings to control the input air velocity. Unless otherwise specified, the furnaces shall be floor mounted, and shall provide a working height above the floor of approximately 30 inches. When required (see 6.2.1), small furnaces shall be designed for bench mounting.

3.2.1 Type I. The type I furnace shall employ the gravity convection method of heating. Convection shall be accomplished by fresh air entering at the bottom of the chamber through a diffusion plate to set up the air flow pattern in the work chamber.

3.2.2 Type II. The type II furnace shall employ the forced convection method of heating. Design shall provide one or more electric motor driven fans in the air flow path to circulate the air through the work chamber.

* 3.2.3 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the furnace. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ASTM-E-380.

* 3.2.3.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of measurements.

* 3.2.4 Reclaimed materials. The furnace may contain reclaimed materials provided such materials will not jeopardize the furnace's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the furnace.

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* 3.2.5 Energy efficiency. The furnace and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.6 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the furnace that present safety hazards. The safety devices shall not interfere with the operation of the furnace. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All furnace parts, components, mechanisms, and assemblies furnished on the furnace, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the furnace itself. Additional safety and health requirements shall be as specified (see 6.2.1).

* 3.2.7 Mercury restriction. The furnace shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

* 3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the furnace.

* 3.2.9 Environmental protection. The furnace shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the furnace shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of installation.

3.2.10 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.11 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or furnace.

3.3 Construction. The furnace shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the furnace to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the furnace. Such processes may be used only for enhancing surface finish and appearance.

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3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

* 3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturers standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the furnace and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

* 3.4 Electrical equipment. The furnace shall be furnished complete with all accessory electrical motors, controls, relays, signal lights, and switches pre-wired where possible for operation upon connection to source of power. All power actuated, manually adjusted, and automatic controls shall operate on a nominal 115 volt, 60 Hz current. A fusible disconnect switch shall be provided to interrupt all electrical power to the furnace. All electrical equipment and enclosures shall conform to NEMA Standard ICS-1.

* 3.4.1 Power source. Unless otherwise specified (see 6.2.1), furnaces shall operate from a power source of 460 volts, 60 Hz, three phase current.

3.4.2 Motors. Motors rated in excess of 1/2 horsepower shall be dual voltage, 230/460 volt, 60 Hz type, and shall be compatible with 3.4.1.

3.4.3 Electrical connections. Electrical connections of conductors and terminal parts shall be of the pressure or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering.

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3.5 Performance. Each furnace shall be capable of operating continuously with a capacity load at any temperature within its operating range. The minimum power input (heater KW rating) shall be not less than the power input requirements specified in table II for each size furnace. If a size different from the dimensions listed in table I is required, the minimum power input requirements shall be as specified (see 6.2.1).

3.5.1 Operating temperature range.

3.5.1.1 Class A. The operating temperature range of the class A furnaces shall be from not higher than a minimum of 200°F through not less than a maximum of 1000°F.

3.5.1.2 Class B. The operating temperature range of the class B furnaces shall be from not higher than a minimum of 200°F through not less than a maximum of 1350°F.

3.5.2 Temperature uniformity.

3.5.2.1 Type I. The type I, class A and class B furnaces, when charged and the temperature stabilized, the temperature variation from chamber control point to any other point within the chamber effective working area shall not vary more than 40°F. The furnaces shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature within its operating range.

3.5.2.2 Type II. The type II, class A and class B furnaces, when charged and the temperature stabilized, the temperature variation from chamber control point in the chamber to any other point within chamber effective working area shall not vary more than 10°F. The furnaces shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature within its operating range. The temperature uniformity of the type II furnaces shall conform to the uniformity requirements of MIL-H-6088, Heat Treatment of Aluminum Alloys.

3.5.3 Load capacity. The load capacity of each type and class furnace shall be not less than 50 pounds per square foot of hearth plate area and 25 pounds per square foot of shelf area, based on two shelves per chamber, for a total capacity of 100 pounds per square foot of chamber area.

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* TABLE II. Power input requirements.

Size	Class A KW Input Not Less Than	Class B KW Input Not Less Than
1	18	20
2	20	25
3	30	34
4	35	40
5	45	50
6	55	65
7	70	90
8	80	100
9	100	120

3.6 Components and systems.

3.6.1 Frame. The furnaces shall have a structural steel frame or comparable structure, and shall provide a stable support for the work chamber and a maximum charge.

3.6.2 Chamber cabinet. The outer shell shall be constructed of aluminized or galvanized corrosion-resistant sheet steel, not less than 0.035 inch in thickness. Unless otherwise specified, the inner walls, including the top and bottom shall be constructed of aluminized steel. When required (see 6.2.1), the inner walls shall be constructed to type 304, 18-8 stainless steel. The inner and outer walls and hearth area shall be reinforced with corrosion-resistant structural angles and shapes, and the front of the chamber shall be reinforced with a steel plate at least 3/8-inch in thickness. The hearth area shall be constructed of 10-gauge or heavier steel plate. Reinforcing shall be provided at points of attachments of door hinges, latches, shelf supports, blower motor and fan mountings. Construction of the chamber shall be sufficient to prevent any visible buckling or bulging when operating at maximum temperature with maximum loads.

3.6.2.1 Shelf supports and shelves. The interior walls shall be provided with shelf supports constructed of the same material as the walls. Unless otherwise specified (see 6.2.1), the supports shall permit the shelves to be placed at intervals of 6 inches, with one shelf furnished for chambers having a height of 12 inches and two shelves furnished for chambers having a height of 18 inches or larger. The shelves shall be of the removable open grid type, constructed of stainless steel, and shall be capable of supporting a workload equivalent to at least 25 pounds per square foot of surface area at maximum furnace temperature.

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* 3.6.3 Insulation. The chamber shall be insulated between the inner and outer shell (including top and bottom) with fireproof industrial rock wool or ceramic fiber material. The insulation shall be nonpowdering nonsettling material and shall not support any load except its own weight. The overall quality and thickness of the insulation shall be such that the outside surface temperature of the chamber will not exceed 80°F above ambient air temperature with the chamber temperature stabilized at maximum temperature.

3.6.4 Door. Unless otherwise specified, the furnaces shall be provided with a single plug type door, hinged on the left side of the furnace as viewed from the front. When required (see 6.2.1), the furnaces shall be provided with a vertical lift door, either manually operated counterbalanced, or air operated as specified. Each door shall overlap the door opening on all four sides at least one inch; and shall be provided with a suitable latch, pull handle, and means for sealing in the closed position. The door shall be insulated and constructed of the same material as the chamber.

3.6.5 Air circulating system. The type II, class A and class B furnaces shall be provided with one or more fans of the circulating type, constructed of heat resistant, nonoxidizing alloy. The fans shall be dynamically balanced and supported on stainless steel shafts with anti-friction bearings. The system shall provide for heated air to enter the chamber in a definite air flow pattern, and flow through the work area in such a way to assure the temperature uniformity specified in 3.5. The capacity of the air system shall be sufficient to provide air changes equivalent to the cubic feet of the heating chamber at least 75 times per minute, with means to adjust air velocity. Fans shall be interlocked with the heating elements, or shall be provided with a safety switch to prevent operation of the furnace in the event of fan failure.

3.6.6 Heating elements. The heating elements shall be designed and located to meet the heating requirements and temperature uniformity specified herein. The element shall be located in an area separated from the work chamber, or shielded from the work chamber to prevent direct radiant heat on the work. The elements shall be fabricated from an alloy containing approximately 35 percent nickel and 20 percent chrome, or from an alloy having as good or better characteristics. The power density on the heating elements shall not exceed 15 watts per square inch of radiation surface. The terminals of the elements shall have at least twice the current carrying capacity of the heating elements.

* 3.6.7 Temperature control system. Each furnace shall be provided with an indicating, automatic temperature control system. The control's temperature indicator shall have a calibrated scale of at least six inches in length with calibration marks indicating increments of 5° Fahrenheit and shall have numbered increments not greater than 200°F apart. The control indicator shall have a range of not less than 100°F above the maximum temperature range of the furnace. Solid state design shall be employed where possible. Unless otherwise specified, the instrument and accessory controls shall be

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mounted in the chamber cabinet, or provided in a separate dustproof enclosure with provisions for mounting to the side of the chamber cabinet. When required (see 6.2.1), the instrument and controls shall be mounted in a separate floor mounted instrument panel conforming to NEMA ICS-1 with 25 feet of thermocouple lead wire having the same thermoelectric characteristics as the thermocouple. Control accuracy shall be in accordance with 3.6.7.1 and 3.6.7.2 for the type furnace specified.

3.6.7.1 Type I. The control system of the type I, class A and class B furnaces shall be capable of maintaining the set control temperature with a deviation of not more than ± one percent of scale range.

3.6.7.2 Type II. The temperature control for the type II, class A and class B furnaces shall be capable of maintaining the set control temperature with a deviation of not more than ± 2 °F. Thermocouple assemblies conforming to ANSI MC96.1 shall be used as the temperature sensing device for each instrument. Accuracy of the temperature control system shall be such that with the chamber temperature at equilibrium, the maximum accumulative error of all elements in the system shall not exceed 20°F between control instrument set point and all points in the work chamber.

3.6.7.3 Temperature recorder. When specified (see 6.2.1), the controls shall be of the indicating, controlling, strip chart recording system, with a chart width of not less than six inches and chart speed of two inches per hour.

3.6.7.4 Program control. When required, a temperature programmer shall be furnished and integrated with the furnace control system to provide heating, soaking and cooling cycles as specified (see 6.2.1).

3.6.7.5 Excess temperature control. When specified (see 6.2.1), the furnaces shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the control circuitry and be accurate within one percent of scale.

3.7 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

* 3.8. Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Inside dimensions (width, length, height)
Temperature range, °F
Contract Number or Order Number
National Stock Number or Plant Equipment Code (NSN Preferred)
Date of manufacture

3.9 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.10 Workmanship. Workmanship of the furnace and accessories shall be commensurate with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.1. Failure of the item to pass the examination or tests shall be cause for rejection.

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4.5 Examination. The furnace and equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

* 4.6 Tests. All tests shall be conducted at an ambient temperature of 77°F ± 18°F. Test instruments and sensing elements used to perform the test specified for the furnace and instruments shall have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy within the previous six months to read accurately within ± 2°F.

4.6.1 Circuit test. Each circuit of the electrical system shall be tested for continuity, short circuits, and faulty ground in accordance with the requirements of NEMA ICS-1 standard.

4.6.2 Operational test. The furnace shall be prepared for operation and operated in accordance with the manufacturer's handbook of instructions until the chamber temperature has stabilized at maximum operating temperature. With the chamber at maximum temperature, the hearth plate and shelves shall be loaded with a maximum charge of iron or steel in accordance with 3.5.3, and the furnace shall continue operating with maximum power input for at least 30 minutes or until the temperature has recovered to maximum. The charge shall consist of various size weights, and the base area of the largest weight shall not exceed 30 square inches. The charge shall not be removed from the chamber, the chamber shall be allowed to cool to a temperature of 77 ± 18°F, and the furnace shall be inspected for damage. Shelf supports, shelves and the hearth plate shall not evidence any buckling, bending, or permanent deformation.

4.6.3 Performance test. The furnace chamber shall be loaded with a batch type charge equivalent to at least 50 percent of maximum load, and a chamber temperature survey performed in the effective work area to determine conformance with 3.5. The effective work area (test zone) is defined as 1/6 less than the actual cubic volume of chamber air space. A minimum of nine test thermocouples shall be used, with one thermocouple located in each corner of the test zone, and one located within six inches of the control thermocouple in the chamber. For furnaces of 10 cubic feet or less, the temperature survey may be performed with a minimum of three thermocouples with one each located at the front, middle, and rear of the test zone. The survey shall be performed under equilibrium conditions at the minimum and maximum temperature range of the furnace. Readings should be continuously recorded or taken frequently enough to determine when the temperature of the test zone approaches the bottom of the temperature range being surveyed. From that time until thermal equilibrium is reached, none of the temperature readings should exceed the maximum of the range being surveyed. After thermal equilibrium is reached, readings shall be taken at not greater than five minute intervals for at least 30 minutes, and the

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maximum temperature variation of all points under test shall not exceed the limits specified in 3.5 for the type furnace under test. With the temperature stabilized at maximum operating temperature, the outside surface shell and door temperature shall be checked at random locations, and shall not exceed the limits specified for the class furnace.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

* 5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D 3951. When required, preservation, and packing shall conform to level A, B, or C requirements of MIL-F-3296 as specified (see 6.2.1), and marking shall be as specified therein.

6. NOTES

6.1 Intended use.

6.1.1 Type I. The type I furnaces are intended for use in laboratory or production heat treating processes where critical temperature control is not required.

6.1.2 Type II. The type II furnaces are intended primarily for use in preheating, stress relieving, forming and heat treatment of steels and alloys such as magnesium, aluminum, and titanium.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.

b. Type and class required (see 1.2.1).

c. Size required (see 1.2.2).

d. First article, when required (see 3.1).

e. Bench mounting, when required (see 3.2).

f. If furnace is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.3).

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g. Dual calibrations (US and SI), if required (see 3.2.3.1).

h. Additional safety and health requirements, if required (see 3.2.6).

i. Painting, if different (see 3.3.5).

j. Power supply, if different (see 3.4.1).

k. Power input requirements, if the dimensions required are different from the sizes listed in table I (see 3.5).

l. Stainless steel inner wall, when required (see 3.6.2).

m. Dimension between shelves and number of shelves to be furnished with each furnace, if different (see 3.6.2.1).

n. Door and method of operating, if different (see 3.6.4).

o. Floor mounted instrument panel, when required (see 3.6.7).

p. Temperature recorder, when required (see 3.6.7.3).

q. Program control and operating characteristics, when required (see 3.6.7.4).

r. Excess temperature controller, when required (see 3.6.7.5).

s. Repair parts, when required (see 3.7).

t. Nameplate, if different (see 3.8).

u. First article inspection, if different (see 4.3).

v. Quality conformance inspection, if different (see 4.4).

w. Level of preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. All data requirements, such as, parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List and shall be furnished in accordance with each specified DD Form 1664, Data Item Description as shown on the DD Form 1423.

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6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.6 requires compliance only with those OSHA requirements that concern the furnace itself. It does not require compliance with those OSHA requirements that concern "the furnace in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits that total hazard level of these hazards in the environment (and does not limit the hazard level of individual furnaces in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional furnace requirements that will integrate the new furnace into its future operating environment. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(i).

6.5 Marginal notation. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document, based on the entire content, irrespective of the marginal notations and relation to the last previous issue.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH
Air Force - 84
DLA - GS

User Activities:
Army - SM
Navy - MC

Preparing Activity:
DLA - IP

Project Number:
3424-0105

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-80188B	Furnaces, Heat Treating, Electric, Metal Lined Box Design

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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